Corrigé — Exercice 15

1) $z_B = \dfrac{2}{\overline{z_A}} = \dfrac{2z_A}{|z_A|^2} = \dfrac{2(1+i\sqrt{3})}{4} = \dfrac{1}{2} + \dfrac{\sqrt{3}}{2}\, i.$

2) $|z_B - 1| = \left| -\dfrac{1}{2} + \dfrac{\sqrt{3}}{2}\, i \right| = 1 \Rightarrow B \in \mathcal{C}(I, 1).$

3a) $OA = |z_A| = 2.$

4) $\mathcal{C} : z\overline{z} = z + \overline{z} \iff x^2 + y^2 = 2x \iff (x-1)^2 + y^2 = 1$: cercle de centre $I(1,0)$, rayon 1.
$\Delta : |z| = |z-2| \iff x = 1$: droite verticale.